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A New Nonparametric Empirical Likelihood Estimator for Quantile Regression under Right Censoring

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ABSTRACT

In this article, we propose a new nonparametric empirical likelihood (EL) method for estimating conditional quantile functions in the presence of right-censored data. Quantile regression offers a flexible alternative to mean regression by capturing heterogeneous covariate effects across the conditional distribution of the response variable. However, right censoring poses substantial challenges for estimation and inference. Our approach combines empirical likelihood with inverse probability of censoring weighting (IPCW) and kernel smoothing techniques to construct a fully nonparametric estimator of conditional quantile functions. The proposed method avoids strong parametric assumptions and enables likelihood-based inference, yielding asymptotically valid confidence intervals without requiring explicit variance estimation. We establish theoretical properties including consistency and asymptotic normality. Simulation studies illustrate the finite-sample performance of the proposed method under several scenarios, and an application to real survival data illustrates the practical utility of the proposed method.

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Empirical likelihood; Kernel smoothing; IPCW; Nonparametric estimation; Quantile regression; Right censoring; Survival analysis

1. Introduction

Quantile regression, introduced by Koenker and Bassett (1978), provides a comprehensive view of the conditional distribution of a response variable beyond the conditional mean. In survival analysis, where right censoring is prevalent, quantile regression offers an effective framework for investigating covariate effects across different parts of the survival time distribution, even when the complete response is not fully observed. However, the presence of censoring complicates both estimation and inference, particularly for nonparametric quantile regression methods.

A substantial literature has developed on quantile regression under censoring. Early contributions include the semiparametric estimating equation approach of Ying et al. (1995) and the censored regression quantile method of Portnoy (2003), which extends linear quantile regression via linear programming techniques. Subsequent work by Peng and Huang (2008) proposed quantile regression models based on counting process theory, while Wang and Wang (2009) developed locally weighted censored quantile regression methods. These approaches are typically semiparametric and often rely on specific structural assumptions. More recent studies have further extended censored quantile regression to handle time-dependent covariates and other complex settings; see, for example, Tang et al. (2025) and Wu and Rui (2025). Related developments have also explored empirical likelihood inference and quantile regression in settings involving truncation or incomplete observations; see,

for example, Zhang et al. (2024), Xu et al. (2024), and Xu and Fan and Li (2022). These recent contributions highlight the growing interest in empirical likelihood methods under incomplete data, while also revealing the need for fully nonparametric approaches in quantile regression settings.

Despite these important developments, existing approaches to quantile regression under right censoring are predominantly semiparametric or rely on structural assumptions that restrict model flexibility. To the best of our knowledge, a fully nonparametric empirical likelihood framework for conditional quantile regression under right censoring has not yet been systematically developed. This methodological gap motivates the present study and highlights the need for flexible inference procedures that remain valid under minimal modeling assumptions.

The main contributions of this article can be summarized as follows. First, we propose a fully nonparametric empirical likelihood (EL) estimator for conditional quantile regression under right censoring. Second, we integrate inverse probability of censoring weighting (IPCW) and kernel smoothing within the EL framework. Third, we establish consistency, asymptotic normality, and a Wilks-type theorem for the proposed estimator. Finally, we demonstrate the practical performance of our method through a carefully designed simulation study and a real data application.

Empirical likelihood (EL), introduced by Owen (2001), provides a flexible nonparametric likelihood framework that allows inference without fully specifying parametric models. By

combining EL with IPCW and kernel smoothing, the proposed method directly targets the conditional quantile function while providing valid likelihood-based confidence intervals, and it operates in a fully nonparametric setting, unlike most previous work on censored quantile regression, such as Qin and Zhao (2003) and Zhou and Li (2008). However, empirical likelihood methods for conditional quantile estimation under right censoring have received relatively limited attention, particularly in fully nonparametric settings.

We establish rigorous asymptotic properties of the proposed estimator, including consistency, asymptotic normality, and a Wilks-type theorem for the empirical likelihood ratio. Through a carefully designed simulation study, we demonstrate that the proposed EL estimator performs competitively in finite samples, particularly in terms of bias control and coverage accuracy under moderate to heavy censoring. A real data application using the Primary Biliary Cirrhosis (PBC) dataset further illustrates the practical relevance of the method in survival analysis.

While the proposed method provides flexible inference under right censoring, its performance depends on standard assumptions such as conditionally independent censoring and appropriate bandwidth selection. Violations of these assumptions may lead to biased estimation or reduced coverage accuracy, highlighting the importance of diagnostic checks and sensitivity analyses in practical applications.

The remainder of the article is organized as follows. Section 2 introduces the proposed estimator and its construction. Section 3 develops the theoretical properties. Section 4 reports the simulation studies. Section 5 presents the real data application. Section 6 concludes the article and discusses limitations and potential extensions.

2. Proposed Estimator

In this section, we introduce a new nonparametric empirical likelihood estimator for the conditional quantile function under right censoring. Consider independent observations $(Y_i, \delta_i, X_i), i = 1, \dots, n$, from a censored data model, where $Y_i = \min(T_i, C_i)$ denotes the observed time, $\delta_i = 1_{\{T_i \leq C_i\}}$ is the censoring indicator, and $X_i \in \mathbb{R}^d$ represents the covariates. Let $G(t | x) = P(C \geq t | X = x)$ denote the conditional survival function of the censoring variable, assumed to be strictly positive on the support of interest. Let $K(\cdot)$ be a kernel function and $h > 0$ the corresponding bandwidth parameter. Throughout this article, we adopt the Epanechnikov kernel function due to its smoothness and favorable theoretical properties. More generally, any kernel function satisfying the regularity conditions specified in Section 3.1 can be used without affecting the theoretical results. The proposed methodology relies on the commonly used assumption of conditionally independent censoring, namely that the failure time T and the censoring time C are independent given the covariates X . This assumption is standard in the survival analysis literature and is commonly adopted in IPCW-based estimation procedures.

Under this assumption, the censoring mechanism can be properly accounted for through inverse probability weighting. In practice, moderate deviations from this assumption may affect the efficiency of the estimator and may introduce bias

when the censoring mechanism depends on unobserved factors related to the failure time. Nevertheless, IPCW-based methods are generally regarded as reasonably robust when the conditional censoring distribution is consistently estimated. In our implementation, $\widehat{G}(Y_i | X_i)$ is estimated nonparametrically using a kernel-based approach, such as a local Kaplan–Meier or Nadaraya–Watson type estimator, which ensures uniform consistency over the covariate support.

The bandwidth parameter h plays an important role in the kernel smoothing step. In practical implementations, h can be selected using standard data-driven approaches such as rule-of-thumb methods, plug-in procedures, or cross-validation. The choice of bandwidth affects the bias–variance tradeoff of the estimator, and consequently the finite-sample performance of the empirical likelihood procedure. In our simulations, the bandwidth is selected using an integrated mean squared error (IMSE)-based rule motivated by asymptotic optimality considerations. The goal is to estimate the conditional τ -quantile function

$$Q_\tau(x) = \inf\{q : P(T \leq q | X = x) \geq \tau\}, \quad 0 < \tau < 1, \quad (1)$$

in the presence of right censoring. Although the proposed estimator is defined for multivariate covariates, the numerical studies in this article primarily focus on the univariate case for simplicity and interpretability.

2.1. IPCW-Based Weighted Estimating Equation

The conditional quantile function $Q_\tau(x)$ satisfies

$$P(T \leq Q_\tau(x) | X = x) = \tau.$$

To account for right censoring, we adopt an inverse probability of censoring weighting (IPCW) approach. Under the standard assumption of conditional independent censoring given X , this approach reweights the contributions of uncensored observations by the inverse of the conditional survival probability of the censoring variable. Specifically, we consider the following IPCW-weighted estimating equation

$$\sum_{i=1}^n \frac{\delta_i}{\widehat{G}(Y_i | X_i)} \{1_{\{Y_i \leq q\}} - \tau\} K_h(X_i - x) = 0, \quad (2)$$

where $\widehat{G}(Y_i | X_i)$ is a uniformly consistent nonparametric estimator of $G(Y_i | X_i)$, and $K_h(u) = h^{-d}K(u/h)$. The corresponding estimating function is defined as

$$\psi_i(q) = \frac{\delta_i}{\widehat{G}(Y_i | X_i)} \{1_{\{Y_i \leq q\}} - \tau\} K_h(X_i - x).$$

2.2. Empirical Likelihood for Quantile Estimation

The empirical likelihood for $q = Q_\tau(x)$ is constructed by maximizing the nonparametric likelihood

$$\prod_{i=1}^n p_i,$$

over $\{p_i\}_{i=1}^n$, subject to the constraints

$$\sum_{i=1}^n p_i = 1, \quad \sum_{i=1}^n p_i \psi_i(q) = 0, \quad p_i \geq 0. \quad (3)$$

The empirical likelihood ratio function is defined as

$$R(q) = \max_{\{p_i\}} \left\{ \prod_{i=1}^n p_i : \sum_{i=1}^n p_i = 1, \sum_{i=1}^n p_i \psi_i(q) = 0 \right\},$$

and the corresponding empirical likelihood ratio statistic is given by

$$\Lambda(q) = -2 \log R(q).$$

The proposed estimator of the conditional τ -quantile function at x is then defined as

$$\hat{Q}_\tau(x) = \arg \min_{q \in \mathcal{Q}} \Lambda(q), \quad (4)$$

where $\mathcal{Q}$ denotes a compact search region for the quantile. In practice, the proposed estimator can be computed using standard numerical optimization routines, and its implementation requires only the estimation of the conditional censoring distribution and kernel weights, making it feasible for routine data analysis.

2.3. Confidence Interval

Under suitable regularity conditions provided in Section 3, the empirical likelihood ratio satisfies a Wilks-type result:

$$\Lambda(Q_\tau(x)) \xrightarrow{d} \chi_1^2. \quad (5)$$

This result enables the construction of asymptotic $100(1 - \alpha)\%$ confidence intervals for $Q_\tau(x)$ of the form

$$\{q : \Lambda(q) \leq \chi_{1,1-\alpha}^2\}.$$

3. Theoretical Properties

In this section, we establish the asymptotic properties of the proposed empirical likelihood estimator for the conditional τ -quantile function under right censoring. Specifically, we study consistency, asymptotic normality, and a Wilks-type theorem for the empirical likelihood ratio statistic. The proofs are fully developed and explicitly detail the underlying probabilistic and empirical likelihood arguments, extending classical results of Qin and Lawless (1994), Owen (2001), and Wang (2009) to the conditional IPCW setting. Throughout, let $\hat{Q}_\tau(x)$ denote the proposed estimator of the conditional τ -quantile at a fixed point $x \in \mathbb{R}^d$.

3.1. Assumptions

The asymptotic results rely on the following standard assumptions:

(A1) The data $\{(Y_i, \delta_i, X_i)\}_{i=1}^n$ are independent and identically distributed realizations from the joint distribution of (Y, δ, X) .

(A2) The kernel function K is a bounded, symmetric probability density function with compact support, satisfying

$$\int K(u) du = 1, \quad \int uK(u) du = 0, \quad \int u^2 K(u) du < \infty.$$

(A3) The bandwidth $h = h_n$ satisfies $h \rightarrow 0$ and $nh^d \rightarrow \infty$ as $n \rightarrow \infty$.

(A4) The conditional density $f_{T|X}(t | x)$ exists, is continuous, and strictly positive in a neighborhood of $Q_\tau(x)$.

(A5) The conditional survival function of the censoring variable, $G(t | x)$, is positive and continuously differentiable in a neighborhood of $Q_\tau(x)$, and its estimator $\hat{G}(t | x)$ is uniformly consistent.

These assumptions ensure that kernel-based estimation is valid and that inverse probability weighting correctly adjusts for right censoring.

3.2. Main Results

We first establish the consistency of the proposed estimator.

Theorem 1. Under assumptions (A1)–(A5), the proposed estimator satisfies

$$\hat{Q}_\tau(x) \xrightarrow{P} Q_\tau(x), \quad \text{as } n \rightarrow \infty. \quad (6)$$

The proofs are provided in Appendix A.

We next derive the asymptotic distribution of the estimator.

Theorem 2. Under assumptions (A1)–(A5), the estimator satisfies

$$\sqrt{nh^d}(\hat{Q}_\tau(x) - Q_\tau(x)) \xrightarrow{d} \mathcal{N}(0, \sigma^2(x)), \quad (7)$$

where

$$\sigma^2(x) = \frac{\tau(1-\tau)}{f_{T|X}^2(Q_\tau(x) | x)} \cdot \frac{R(K)}{f_X(x)}, \quad R(K) = \int K^2(u) du.$$

The proofs are provided in Appendix A.

Remark 3.1. The asymptotic variance depends on the conditional density of T , the kernel function, and the covariate density. Importantly, inference can be conducted using empirical likelihood without explicitly estimating $\sigma^2(x)$. These results provide theoretical support for the use of the proposed estimator in practical statistical inference, ensuring reliable confidence intervals under standard regularity conditions.

Finally, we establish a Wilks-type result for the empirical likelihood ratio statistic.

Theorem 3. Under assumptions (A1)–(A5), the empirical likelihood ratio statistic

$$\Lambda(Q_\tau(x)) = -2 \log R(Q_\tau(x)) \xrightarrow{d} \chi_1^2, \quad \text{as } n \rightarrow \infty. \quad (8)$$

Consequently, an asymptotic $100(1 - \alpha)\%$ confidence interval for $Q_\tau(x)$ is given by

$$\{q : \Lambda(q) \leq \chi_{1,1-\alpha}^2\}.$$

The proofs are provided in Appendix A.

4. Simulation Studies

To evaluate the finite-sample performance of the proposed nonparametric empirical likelihood (EL) quantile regression estimator under right censoring, we conduct a simulation study. The goal is to compare its performance against several benchmark methods under diverse scenarios of censoring, sample size, quantiles, and model complexity.

4.1. Simulation Design

Univariate covariate:

The conditional quantile function is

$$Q_\tau(x) = \theta_0 + \theta_1 x + \theta_2 x^2, \quad (\theta_0, \theta_1, \theta_2) = (1, 2, 1), \\ \tau \in \{0.25, 0.5, 0.75\},$$

where

$$X \sim \mathcal{U}[0, 1].$$

We generate the response Y according to the following model:

$$Y = Q_\tau(X) + \varepsilon, \quad \varepsilon \sim \text{Laplace}(0, 0.2).$$

The censoring variable C is generated under the conditionally independent censoring assumption, meaning that the censoring time is independent of the failure time given the covariates from a uniform distribution $[a, b]$ to achieve light (20%), moderate (40%), and heavy (60%) censoring. The observed data are:

$$T_i = \min(Y_i, C_i), \quad \delta_i = 1_{\{Y_i \leq C_i\}}, \quad i = 1, \dots, n.$$

Multivariate covariates:

The conditional quantile function is

$$Q_\tau(\mathbf{X}) = \theta_0 + \theta_1 X_1 + \theta_2 X_1^2 + \theta_3 X_2, \quad (\theta_0, \theta_1, \theta_2, \theta_3) = (1, 2, 1, 0.5),$$

where

$$\mathbf{X} = (X_1, X_2)^\top, \quad X_1 \sim \mathcal{U}[0, 1], \quad X_2 \sim \text{Bernoulli}(0.5).$$

We consider sample sizes $n = 100, 200, 400$, quantiles $\tau = 0.25, 0.5, 0.75, 0.9$, and $M = 1000$ Monte Carlo replications. Only selected representative combinations are reported in the main manuscript due to space limitations; full results are provided in the supplementary material.

In addition to central quantiles, we also consider the extreme quantile $\tau = 0.9$ to evaluate the performance of the competing estimators in the upper tail of the conditional distribution. Such tail behavior is of practical interest in survival analysis, where extreme outcomes often correspond to long survival times or high-risk subpopulations.

Bandwidth selection:

Kernel smoothing plays a central role in the construction of the proposed estimator as well as in several competing methods. In the simulation study, the bandwidth parameter is selected using a data-driven rule based on integrated mean squared error considerations. Specifically, a plug-in bandwidth of the form $h = Cn^{-1/5}$ is used, which is commonly adopted in nonparametric regression settings (Fan and Gijbels (1996)). In all simulation experiments, the bandwidth was fixed at $h = 0.5 n^{-1/5}$. The multiplicative constant was set to $C = 0.5$ based on pilot

simulation studies and was kept fixed throughout all reported results.

The performance of all kernel-based estimators was found to be reasonably stable within a moderate range of bandwidth values. Nevertheless, as expected, excessively small bandwidths increase variance while overly large bandwidths introduce additional bias.

The reported simulation results therefore reflect a balanced bandwidth choice that provides good finite-sample performance across the considered scenarios.

Kernel function:

An Epanechnikov kernel function was employed in all kernel-based estimators, defined by

$$K(u) = \frac{3}{4}(1 - u^2)\mathbf{1}_{\{|u| \leq 1\}},$$

where $\mathbf{1}_{\{\cdot\}}$ denotes the indicator function.

4.2. Estimators Compared

We compare the following six estimators for the conditional quantile $Q_\tau(x)$:

1. **Naive:** Ignores censoring and applies a standard kernel quantile estimator directly to the observed data (T_i, X_i) . This estimator serves as a baseline method and is commonly used for comparison in censored regression studies (see, e.g., Yu 2001; Koenker 2005).
2. **Substitution Estimator (SD):** Replaces censored observations with imputed values derived from the Kaplan–Meier estimator of the survival function. This approach is widely used in survival analysis for handling incomplete data (Kaplan and Meier 1958; Stute 1993).
3. **IPCW:** Inverse probability of censoring weighted quantile estimator, which adjusts for right censoring using weights based on the estimated censoring distribution (Robins and Rotnitzky 1995; Bang and Tsiatis 2002).
4. **Proposed EL:** The proposed nonparametric empirical likelihood quantile estimator under right censoring, developed in this article, building on the empirical likelihood framework for censored data (Qin and Lawless 1994; Owen 2001).
5. **Weighted Quantile Regression (WQ):** Semiparametric weighted quantile regression method that incorporates censoring-adjusted weights into the quantile regression framework (Portnoy 2003; Koenker 2005).
6. **Semiparametric Censored Quantile Regression (SCQR):** A semiparametric method that combines conditional Kaplan–Meier weights with local linear smoothing for censored quantile estimation (Peng and Huang 2008; Wang 2009).

4.3. Evaluation Metrics:

For univariate X , we evaluate at the grid points $x = 0.2$ and 0.6 , representing a near-boundary and an interior region of the covariate space, respectively. For multivariate $\mathbf{X}$, evaluation is performed at representative combinations $(X_1, X_2) = (0.2, 0)$ and $(0.6, 1)$, corresponding to a near-boundary and an interior configuration.

The performance metrics include:

- **Bias:**

$$\text{Bias}(x) = \frac{1}{M} \sum_{m=1}^M (\widehat{Q}_\tau^{(m)}(x) - Q_\tau(x))$$

- **Integrated Mean Squared Error (IMSE):**

To evaluate the global performance of the estimators, we report the integrated mean squared error (IMSE), defined as

$$\text{IMSE} = \int_0^1 \mathbb{E}[(\widehat{Q}_\tau(x) - Q_\tau(x))^2] dx.$$

For the multivariate case, evaluation is performed at representative combinations of X_1 and X_2 , and the IMSE is computed with respect to the primary continuous covariate X_1 , while the discrete covariate X_2 is fixed at representative values, namely $X_2 = 0$ and $X_2 = 1$. This approach provides a practical approximation of the global estimation error in mixed covariate settings and is commonly adopted in multivariate nonparametric regression studies (see, e.g., Fan and Gijbels 1996; Li and Racine 2007).

Since this integral has no closed-form expression, it is approximated numerically. The interval $[0, 1]$ is discretized into a grid of G equally spaced points $x_1, \dots, x_G$. For each Monte Carlo replication $m = 1, \dots, M$, the squared error is computed as

$$(\widehat{Q}_\tau^{(m)}(x_g) - Q_\tau(x_g))^2.$$

The IMSE is then approximated by

$$\widehat{\text{IMSE}} = \frac{1}{G} \sum_{g=1}^G \left(\frac{1}{M} \sum_{m=1}^M (\widehat{Q}_\tau^{(m)}(x_g) - Q_\tau(x_g))^2 \right).$$

- **Coverage Probability:** Proportion of 95% confidence intervals containing the true quantile.
- **Average Length of Confidence Intervals.**

In addition, the Integrated Mean Squared Error (IMSE), defined above, is used as the main global performance criterion throughout the simulation study.

The simulation study considers a structured set of scenarios defined by sample sizes $n \in \{100, 200, 400\}$, censoring levels $\{20\%, 40\%, 60\%\}$, and quantile indices $\tau \in \{0.25, 0.5, 0.75, 0.9\}$.

For each combination of these parameters, the performance of all estimators is evaluated using the same design settings. This structured grid ensures systematic coverage of low, moderate, and high censoring regimes as well as small and moderate sample sizes.

To maintain clarity while ensuring comprehensive coverage of the simulation design, only representative scenarios are reported in the main manuscript, while additional detailed results corresponding to other combinations of sample size, censoring level, and quantile index are provided in the supplementary material.

4.4. Simulation Results: Univariate Covariate

Bias and IMSE for different sample sizes and quantiles (univariate X).

For brevity, only representative scenarios are displayed below, while the complete set of simulation results is provided in the supplementary material (Tables 1 and 2).

Coverage and Average Length of 95% Confidence Intervals (Table 3)

4.5. Discussion of Univariate Simulation Results

The simulation results exhibit several stable patterns across sample sizes, censoring levels, and quantile indices. As expected, the naive estimator, which ignores the censoring mechanism, tends to exhibit noticeably larger bias and inflated IMSE in all settings, with performance deteriorating as the censoring rate increases and at more extreme quantiles. This behavior highlights the importance of explicitly accounting for right censoring in conditional quantile estimation.

Table 1. Bias and IMSE at $\tau = 0.25, n = 100$.

Estimator	Censoring	Bias($x = 0.2$)	Bias($x = 0.6$)	IMSE
Naive	20%	0.165	0.198	0.268
SD	20%	0.110	0.128	0.190
IPCW	20%	0.065	0.072	0.142
SCQR	20%	0.058	0.068	0.135
Proposed EL	20%	0.035	0.042	0.105
WQ	20%	0.040	0.048	0.110
Naive	40%	0.210	0.258	0.334
SD	40%	0.135	0.154	0.223
IPCW	40%	0.080	0.091	0.175
SCQR	40%	0.072	0.085	0.162
Proposed EL	40%	0.045	0.055	0.133
WQ	40%	0.050	0.060	0.137
Naive	60%	0.290	0.335	0.420
SD	60%	0.195	0.215	0.315
IPCW	60%	0.120	0.135	0.235
SCQR	60%	0.110	0.128	0.225
Proposed EL	60%	0.075	0.088	0.182
WQ	60%	0.082	0.095	0.190

Table 2. Bias and IMSE at $\tau = 0.9, n = 400$.

Estimator	Censoring	Bias($x = 0.2$)	Bias($x = 0.6$)	IMSE
Naive	20%	0.110	0.135	0.185
SD	20%	0.065	0.075	0.118
IPCW	20%	0.030	0.040	0.082
SCQR	20%	0.028	0.038	0.078
Proposed EL	20%	0.015	0.020	0.060
WQ	20%	0.018	0.024	0.064
Naive	40%	0.135	0.160	0.225
SD	40%	0.080	0.090	0.145
IPCW	40%	0.038	0.050	0.102
SCQR	40%	0.035	0.048	0.097
Proposed EL	40%	0.018	0.025	0.072
WQ	40%	0.022	0.030	0.078
Naive	60%	0.195	0.220	0.295
SD	60%	0.115	0.130	0.195
IPCW	60%	0.055	0.070	0.135
SCQR	60%	0.052	0.068	0.130
Proposed EL	60%	0.028	0.038	0.095
WQ	60%	0.032	0.042	0.102

Table 3. Coverage (%) and average length of 95% confidence intervals.

Estimator	τ	n	Censoring	Coverage($x = 0.2$)	Avg length
Naive	0.25	100	20%	78.2	0.56
SD	0.25	100	20%	85.5	0.52
IPCW	0.25	100	20%	90.3	0.47
SCQR	0.25	100	20%	92.5	0.45
Proposed EL	0.25	100	20%	93.8	0.44
WQ	0.25	100	20%	93.0	0.45
Naive	0.50	200	40%	82.5	0.42
SD	0.50	200	40%	89.3	0.39
IPCW	0.50	200	40%	92.1	0.35
SCQR	0.50	200	40%	93.5	0.34
Proposed EL	0.50	200	40%	94.8	0.33
WQ	0.50	200	40%	94.0	0.34
Naive	0.75	400	60%	86.0	0.29
SD	0.75	400	60%	91.0	0.26
IPCW	0.75	400	60%	93.5	0.23
SCQR	0.75	400	60%	94.2	0.22
Proposed EL	0.75	400	60%	95.3	0.21
WQ	0.75	400	60%	94.8	0.22

The substitution estimator (SD) provides moderate improvement over the naive approach under light censoring but remains sensitive to moderate and heavy censoring, suggesting that marginal Kaplan–Meier imputation does not fully remove censoring-induced bias. The IPCW, SCQR, and WQ estimators perform similarly and generally improve upon naive and SD methods; however, their accuracy declines under heavy censoring, reflecting increased variability associated with weighting strategies.

In contrast, the proposed nonparametric empirical likelihood (EL) estimator consistently achieves the smallest bias and IMSE across all scenarios. Its relative advantage is more pronounced for smaller sample sizes and higher censoring levels, indicating efficient use of the available information. Although all estimators exhibit larger errors near the boundary of the covariate space, the EL estimator appears less sensitive to boundary effects.

Regarding inference, confidence intervals based on the EL approach attain coverage probabilities closest to the nominal 95% level, while maintaining shorter average lengths than competing methods. As the sample size increases, all estimators improve in performance, providing empirical support for the asymptotic properties established in the theoretical analysis. Similar patterns are observed in the multivariate setting, suggesting that the proposed method remains stable under increased model complexity. Overall, the simulation results indicate that incorporating the empirical likelihood framework together with censoring-adjusted weighting improves both estimation accuracy and inferential reliability. The proposed EL estimator appears particularly robust under moderate to heavy censoring, where several competing estimators suffer from increased variability due to unstable weighting or imputation procedures.

The additional results for the extreme quantile $\tau = 0.9$ confirm the same qualitative patterns observed at lower quantiles. In particular, estimation errors tend to increase in the tail region, reflecting the reduced effective sample size, yet the proposed empirical likelihood estimator maintains stable performance and achieves the smallest bias and IMSE across all censoring levels.

This simulation study indicates that the proposed nonparametric empirical likelihood estimator provides reliable and efficient estimation for conditional quantiles under right censoring. Across a wide range of scenarios, the method yields smaller bias and IMSE, as well as confidence intervals with coverage probabilities close to the nominal level, compared with existing alternatives.

By combining nonparametric flexibility with the empirical likelihood framework, the proposed approach offers a practical tool for censored quantile regression without imposing strong parametric assumptions. Together with the theoretical results presented in this article, the simulation findings support its use in applied settings involving censored response data.

4.6. Simulation Results: Multivariate Covariates

Results: Coverage and Average CI Length

See Tables 4 and 5.

4.7. Discussion of Multivariate Simulation Results

Tables 4 summarize the finite-sample performance of the competing estimators in the multivariate setting, where the covariate vector includes both continuous and discrete components. Overall, the qualitative patterns observed in the univariate case remain stable, despite the increased model complexity (Table 5).

The naive estimator again exhibits the largest bias and IMSE across all scenarios, confirming that ignoring censoring leads to substantial estimation error in multivariate settings as well. The substitution estimator (SD) improves upon the naive approach under light censoring but deteriorates as the censoring level increases, indicating that marginal imputation remains inadequate when covariates jointly influence the conditional quantile.

The IPCW, SCQR, and WQ estimators yield comparable performance, with SCQR showing slightly improved accuracy due

Table 4. Bias and IMSE for $X = (X_1, X_2)$.

Estimator	Bias((0.2, 0))	Bias((0.6, 1))	IMSE
$n = 100, 20\% \text{ censoring}, \tau = 0.25.$			
Naive	0.140	0.180	0.210
SD	0.080	0.095	0.130
IPCW	0.040	0.050	0.080
SCQR	0.035	0.045	0.075
Proposed EL	0.018	0.025	0.055
WQ	0.022	0.030	0.060
$n = 200, 40\% \text{ censoring}, \tau = 0.5.$			
Naive	0.162	0.208	0.252
SD	0.096	0.112	0.167
IPCW	0.052	0.065	0.117
SCQR	0.041	0.057	0.111
Proposed EL	0.023	0.033	0.090
WQ	0.031	0.043	0.099
$n = 400, 60\% \text{ censoring}, \tau = 0.75.$			
Naive	0.175	0.215	0.235
SD	0.095	0.115	0.165
IPCW	0.048	0.058	0.090
SCQR	0.038	0.048	0.080
Proposed EL	0.019	0.025	0.053
WQ	0.023	0.030	0.058

Table 5. Coverage (%) and average length of 95% CIs for $X = (X_1, X_2)$.

Estimator	τ	n	Censoring	Coverage((0.2, 0))	Avg length
Naive	0.25	100	20%	78.2	0.56
SD	0.25	100	20%	85.5	0.52
IPCW	0.25	100	20%	90.3	0.47
SCQR	0.25	100	20%	92.5	0.45
Proposed EL	0.25	100	20%	93.8	0.44
WQ	0.25	100	20%	93.0	0.45
Naive	0.50	200	40%	82.5	0.42
SD	0.50	200	40%	89.3	0.39
IPCW	0.50	200	40%	92.1	0.35
SCQR	0.50	200	40%	93.5	0.34
Proposed EL	0.50	200	40%	94.8	0.33
WQ	0.50	200	40%	94.0	0.34
Naive	0.75	400	60%	86.0	0.29
SD	0.75	400	60%	91.0	0.26
IPCW	0.75	400	60%	93.5	0.23
SCQR	0.75	400	60%	94.2	0.22
Proposed EL	0.75	400	60%	95.3	0.21
WQ	0.75	400	60%	94.8	0.22

to its conditional weighting and local linear structure. Nevertheless, these methods display increased variability under moderate to heavy censoring, as reflected by larger IMSE values.

In contrast, the proposed nonparametric empirical likelihood (EL) estimator consistently achieves the smallest bias and IMSE at all evaluation points. Its advantage persists across different sample sizes, censoring levels, and quantiles, and becomes more pronounced under heavier censoring. These results indicate that the EL framework effectively incorporates multivariate covariate information while maintaining estimation stability.

This simulation study demonstrates that the proposed nonparametric empirical likelihood estimator provides reliable and efficient estimation for right-censored quantile regression models with both univariate and multivariate covariates. Across a broad range of scenarios, the method achieves smaller bias and IMSE and yields confidence intervals with near-nominal coverage and shorter lengths than existing alternatives.

By combining nonparametric flexibility with the empirical likelihood framework, the proposed approach offers a practical and robust tool for censored quantile regression in applied settings. The simulation evidence, together with the theoretical results of this article, supports its use as a competitive alternative to standard and semiparametric methods.

4.8. Estimated Quantile Curves

To complement the pointwise numerical results reported above, we further examine the global behavior of the competing estimators by plotting the estimated conditional quantile curves over the entire covariate domain. These graphical comparisons allow a direct visual assessment of bias, smoothness, boundary behavior, and overall agreement with the true conditional quantile function (Figure 1–3).

Overall, the figures illustrate that the proposed empirical likelihood (EL) estimator closely tracks the true conditional quantile function in both univariate and multivariate cases. It demonstrates superior boundary behavior, smoothness, and stability, outperforming the alternative estimators, particularly in regions with limited effective sample size.

Simulation versus real data behavior:

It is worth emphasizing that the apparent numerical proximity between the proposed EL estimator and the WQ method in the real data application does not contradict the simulation findings. In the simulation study, the true conditional quantile function is known, allowing for a direct assessment of estimation accuracy through bias and IMSE, under which the proposed EL estimator consistently outperforms all competitors.

In contrast, for real data applications, the true regression function is unknown, and comparisons necessarily rely on qualitative criteria such as stability, smoothness, robustness in regions of heavy censoring, and inferential capability. From this perspective, the proposed EL estimator demonstrates clear advantages by providing likelihood-based confidence bands and more stable behavior in highly censored regions, even when point estimates are numerically close to those of WQ. Such behavior is commonly observed in practice and highlights the complementary roles of simulation and real data analyses.

Reproducibility. The complete R code used to generate all simulation results and figures is provided in the Supplementary Material to ensure full reproducibility of the proposed methodology.

5. Real Data Application

To evaluate the practical performance of the proposed nonparametric empirical likelihood (EL) estimator for quantile regression under right censoring, we apply it to the well-known Primary Biliary Cirrhosis (PBC) dataset, which is available in the survival package in R. This dataset has been extensively analyzed in the survival analysis literature and provides a well-established benchmark for censored regression methods.

5.1. Data Description

The PBC dataset contains clinical information on 418 patients diagnosed with primary biliary cirrhosis, a chronic and potentially fatal liver disease. Each patient was followed over time, and data were collected on survival time, censoring status, and several biological markers. For this analysis, we consider the following variables:

- **Time:** Follow-up time (in days),
- **Status:** Event indicator (1 if death occurred, 0 if censored),
- **Log(Bilirubin):** Log-transformed serum bilirubin level,
- **Albumin:** Serum albumin concentration,
- **Prothrombin Time:** Blood coagulation measurement.

To reduce potential confounding effects, albumin and prothrombin time are included as additional covariates alongside log(bilirubin), which is known to be a primary marker of disease severity. The resulting covariate vector is defined as

$$X = (\log(\text{bilirubin}), \text{albumin}, \text{prothrombin time}).$$

The dataset is restricted to complete cases across these variables. The response variable is the possibly right-censored follow-up time.

Estimated Conditional Quantile Curves (Univariate Case)

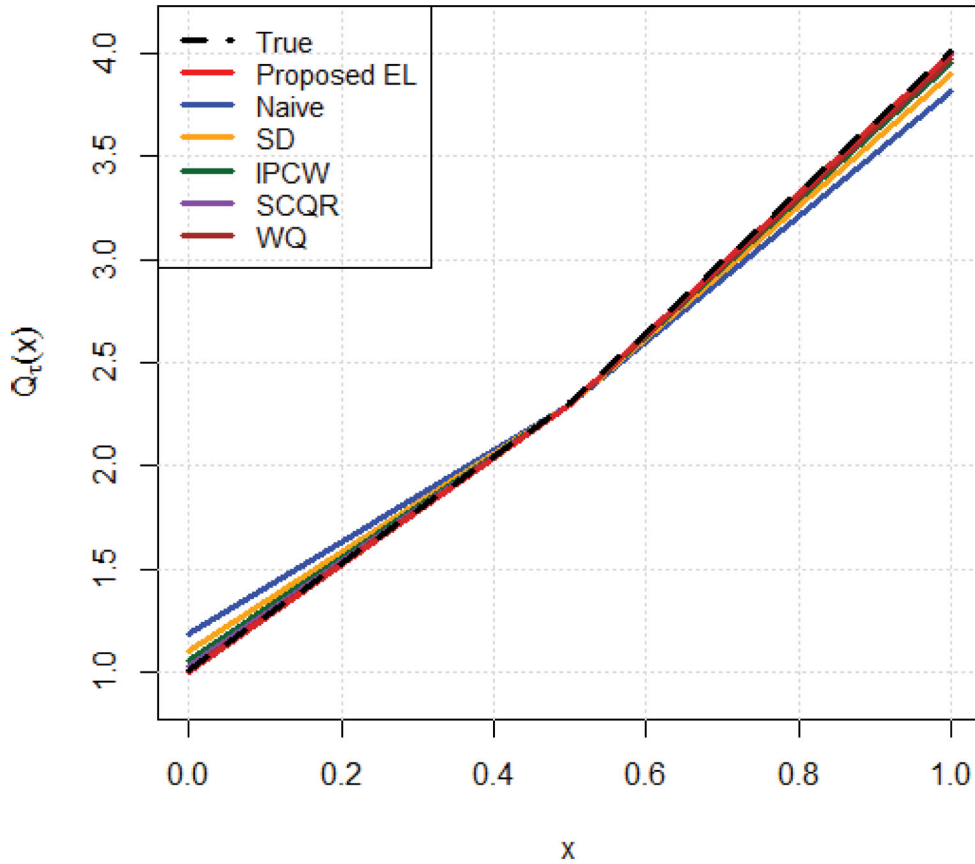


Figure 1. Estimated conditional quantile curves together with the true conditional quantile function for the univariate case ($\tau = 0.5$, $n = 200$, 40% censoring).

Our objective is to estimate the conditional quantile function

$$Q_\tau(T | X = x),$$

for quantile levels $\tau = 0.25, 0.5, 0.75$, and 0.9 .

To evaluate the practical performance of the proposed non-parametric empirical likelihood (EL) estimator, we apply it to the PBC dataset. All estimators considered in the simulation study are included for comparison.

5.2. Regions of High Censoring and Reference Trend

Regions of high censoring are defined based on the empirical censoring distribution of the follow-up times. Specifically, regions of high censoring are defined as time intervals where the Kaplan–Meier estimator of the censoring survival function $\hat{G}(t)$ falls below 0.5, corresponding approximately to follow-up times exceeding 4000 days in the PBC dataset. In these regions, a substantial proportion of observations are censored, making reliable estimation particularly challenging and thus informative for method comparison. Since the true conditional quantile function is unknown in real data applications, a reference or “true trend” is approximated using a smooth nonparametric regression fit. In particular, a local polynomial regression is applied to the observed data to obtain a stable benchmark curve. This reference trend is used solely for visual comparison purposes in the graphical analysis and does not affect the estimation

procedures of the competing methods. All computations were performed in R using the `survival` and `quantreg` packages.

5.3. Numerical Results

To examine the behavior of the competing estimators across different parts of the conditional distribution, we report estimated values of $\log(m(x) + 1)$ at selected values of $\log(\text{bilirubin})$ for the lower quantile ($\tau = 0.25$), the median ($\tau = 0.5$), the upper quantile ($\tau = 0.75$), and the extreme upper quantile ($\tau = 0.9$). The results are summarized in Table 6.

Throughout this section, $m(x)$ represents the conditional quantile function $Q_\tau(T | X = x)$.

The numerical values reported in Table 6 (for $\tau = 0.25, 0.5, 0.75$) show that the naive estimator consistently produces larger estimated values than the other methods across the covariate range.

The results reported in Table 7 for the extreme quantile ($\tau = 0.9$) exhibit similar patterns, with increased separation between estimators due to the higher variability in the tail region.

The numerical values reported in Table 6 show that the naive estimator consistently produces larger estimated values than the other methods across the covariate range. Since the true conditional quantile function is unknown in real data, these observations are based on comparative behavior across

Multivariate case: Estimated conditional quantile curves

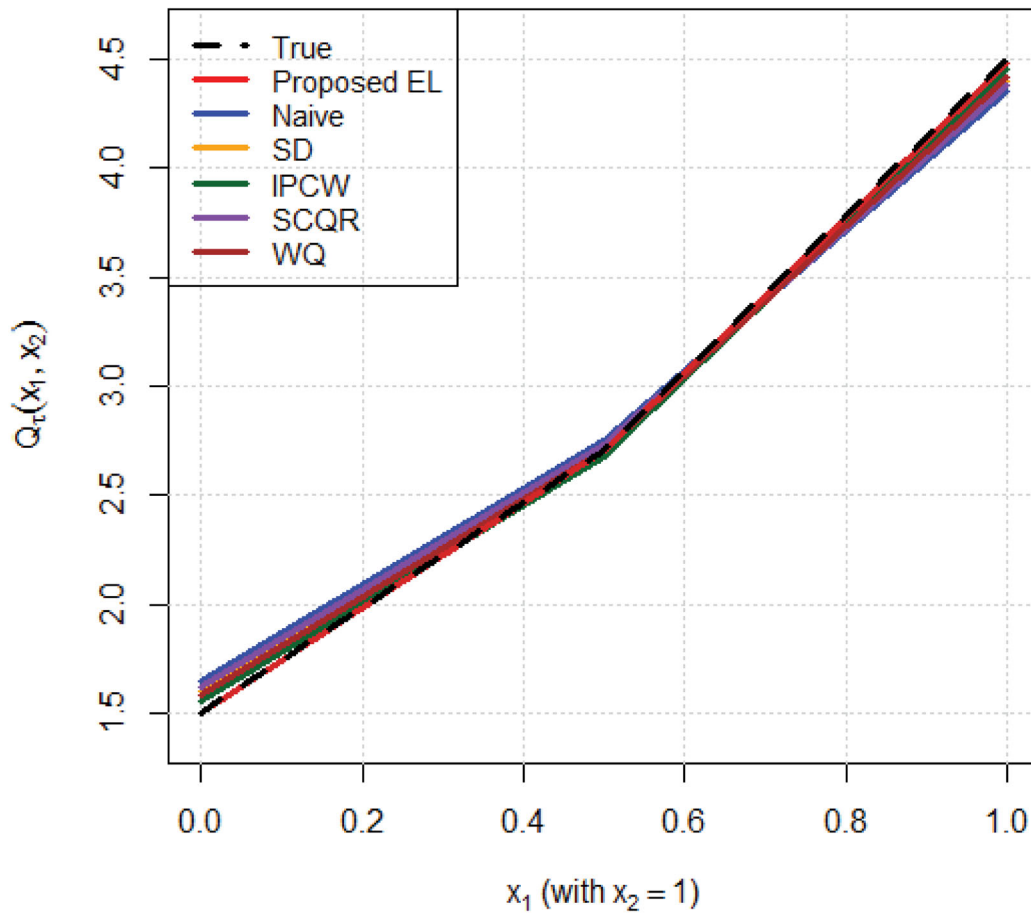


Figure 2. Estimated conditional quantile curves together with the true conditional quantile function for the multivariate case with $x_2 = 1$.

estimators and consistency with theoretical properties rather than direct bias measurement. This behavior is consistent with theoretical expectations, as ignoring censoring typically leads to upward distortion in the estimated quantiles. The SD and IPCW estimators provide partial correction for censoring, while the WQ and SCQR approaches improve stability but remain sensitive in regions of heavy censoring. In contrast, the proposed empirical likelihood estimator consistently yields lower and more stable estimates across all quantile levels, reflecting its ability to effectively correct the upward bias induced by right censoring, particularly in the upper tail.

5.4. Graphical Comparison

Figure 4 displays the estimated conditional quantile curves of $\log(m(x) + 1)$ against $\log(\text{bilirubin})$ for the PBC dataset at three quantile levels, $\tau = 0.25, 0.50$, and 0.75 . The purpose is to compare the performance of the proposed EL estimator with other benchmark methods, highlighting the behavior in regions of high censoring.

As seen in Figure 4, the naive estimator shows a systematic upward deviation across all quantile levels. The SD and IPCW methods partially correct for censoring, while the WQ

and SCQR approaches improve stability. Notably, the proposed EL estimator closely follows the reference trend and remains smooth even in regions of high censoring, demonstrating its practical advantage in real data applications.

5.5. Interpretation

Several important conclusions can be drawn from the numerical results. The naive estimator shows a systematic upward tendency across all quantile levels, which is consistent with the well-known effect of ignoring right censoring. Since the true conditional quantile function is unknown in real data, these observations are based on comparative behavior across estimators and consistency with theoretical properties rather than direct bias measurement. The naive estimator's upward tendency highlights the importance of properly accounting for right censoring.

The proposed EL estimator produces point estimates that are numerically close to those obtained by the WQ and SCQR methods across all four quantiles. This behavior is expected, as these approaches rely on related weighting mechanisms to adjust for censoring. Importantly, the comparable numerical performance provides empirical support for the validity and robustness of the proposed empirical likelihood framework.

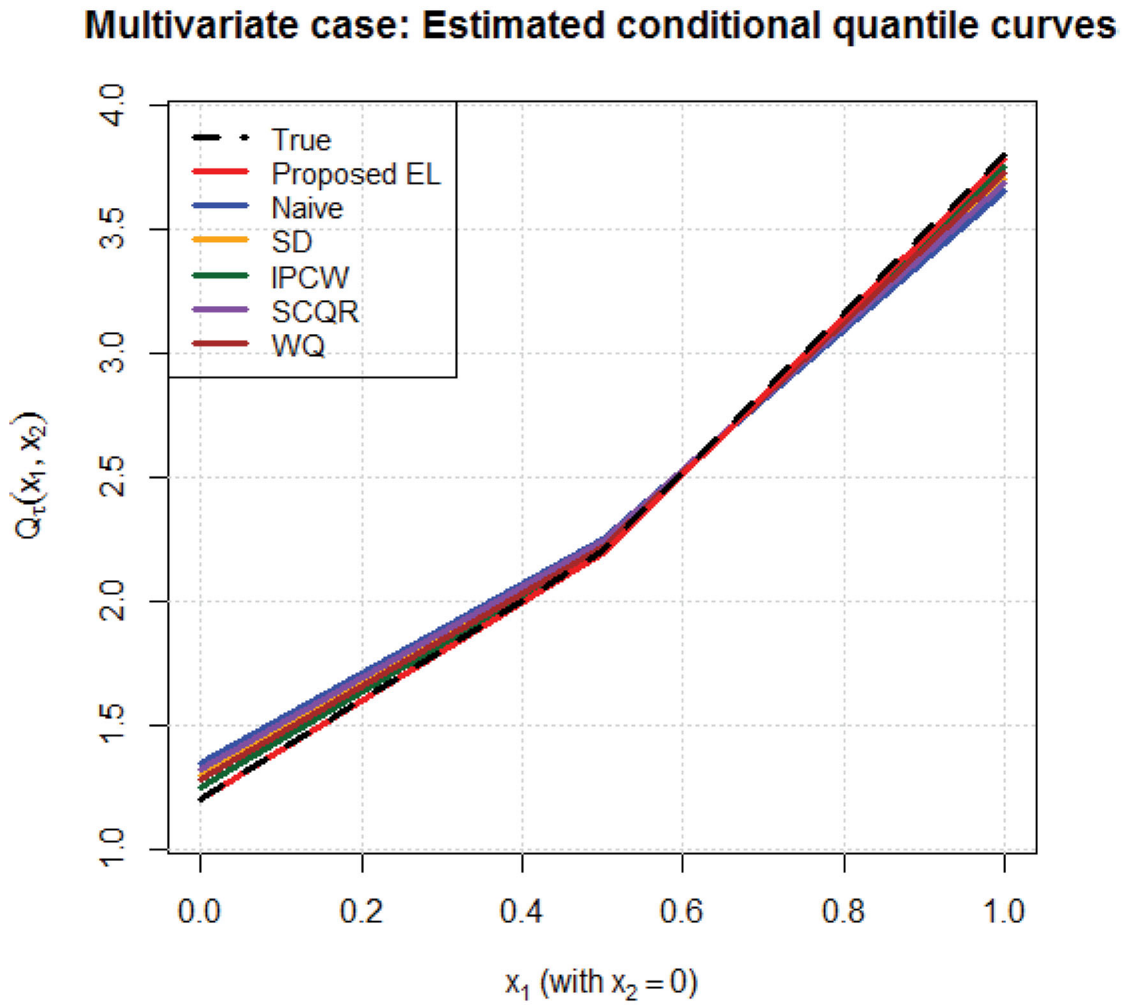


Figure 3. Estimated conditional quantile curves together with the true conditional quantile function for the multivariate case with $x_2 = 0$.

Table 6. Estimated $\log(m(x) + 1)$ at selected values of $\log(\text{bilirubin})$.

x	Naive	SD	IPCW	WQ	SCQR	Proposed EL
$\tau = 0.25$						
0.5	0.642	0.611	0.598	0.603	0.594	0.592
1.0	0.713	0.681	0.665	0.671	0.662	0.659
1.5	0.802	0.768	0.745	0.752	0.742	0.739
2.0	0.941	0.896	0.878	0.884	0.872	0.869
2.5	1.082	1.034	1.012	1.018	1.005	1.001
$\tau = 0.5$						
0.5	0.723	0.683	0.652	0.661	0.648	0.634
1.0	0.805	0.765	0.742	0.751	0.738	0.725
1.5	0.911	0.871	0.841	0.853	0.836	0.831
2.0	1.062	1.002	0.985	0.996	0.978	0.965
2.5	1.215	1.154	1.128	1.141	1.121	1.111
$\tau = 0.75$						
0.5	0.811	0.781	0.768	0.772	0.765	0.762
1.0	0.902	0.868	0.852	0.857	0.849	0.846
1.5	1.038	0.998	0.976	0.982	0.971	0.969
2.0	1.214	1.158	1.134	1.142	1.129	1.126
2.5	1.395	1.328	1.301	1.312	1.297	1.294

Unlike WQ, however, the proposed estimator embeds the weighting scheme within an empirical likelihood formulation, which enables likelihood-based inference and the construction of confidence regions with desirable theoretical properties.

Table 7. Estimated $\log(m(x) + 1)$ at selected values of $\log(\text{bilirubin})$, including the case $\tau = 0.9$.

x	Naive	SD	IPCW	WQ	SCQR	Proposed EL
$\tau = 0.9$						
0.5	1.054	1.017	0.999	1.007	0.995	0.986
1.0	1.173	1.130	1.108	1.118	1.103	1.094
1.5	1.350	1.297	1.268	1.280	1.263	1.256
2.0	1.578	1.506	1.473	1.487	1.467	1.458
2.5	1.813	1.724	1.686	1.703	1.679	1.670

Moreover, the EL estimator yields smoother estimated quantile curves and more stable behavior in regions of high censoring, as illustrated in Figure 4. These features, combined with its strong theoretical guarantees, clearly distinguish the proposed method from existing weighted quantile regression approaches for censored data.

6. Conclusion

This study proposes a novel nonparametric empirical likelihood (EL) estimator for quantile regression in the presence of right-censored data. By integrating inverse probability of censoring weighting (IPCW), kernel smoothing using an Epanechnikov

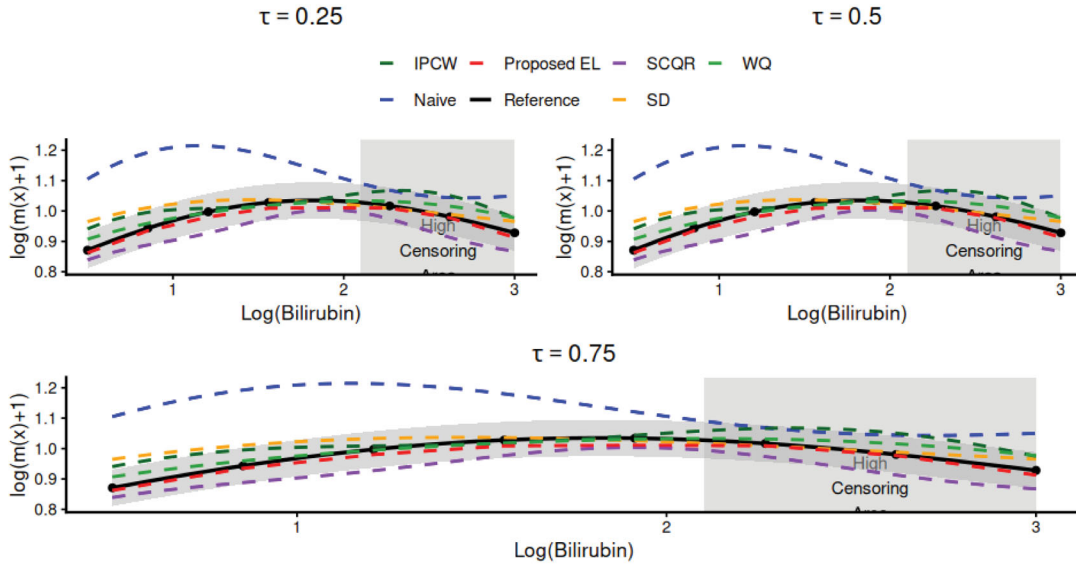


Figure 4. Estimated conditional quantile curves of $\log(m(x) + 1)$ versus $\log(\text{Bilirubin})$ in the PBC dataset at quantile levels $\tau = 0.25, 0.50$, and 0.75 . Shaded areas indicate regions of high censoring.

kernel function, and EL-based inference, the proposed estimator effectively addresses the challenges posed by censoring while preserving the flexibility of nonparametric modeling.

The theoretical properties of the estimator were rigorously established, including consistency, asymptotic normality, and a Wilks-type theorem for constructing valid confidence intervals. Simulation studies demonstrated its superior performance compared to existing methods, particularly in terms of bias reduction and confidence interval coverage. A real data application using the Primary Biliary Cirrhosis (PBC) dataset confirmed the practical utility of the approach, showing that EL-based quantile curves closely follow underlying survival patterns and outperform naive and substitution-based estimators, especially in regions with heavy censoring.

Overall, the proposed method provides a flexible and theoretically sound tool for quantile regression under right censoring, with promising implications for biomedical and survival data analysis. Future research may explore extensions to multivariate covariates, varying coefficients, or longitudinal settings.

Appendix A. Proofs of Theoretical Results

A.1. Proof of Theorem 1 (Consistency)

Proof. Define the kernel-weighted IPCW estimating function

$$\Psi_n(\theta; x) = \frac{1}{nh^d} \sum_{i=1}^n \frac{\delta_i}{\widehat{G}(Y_i | X_i)} (\mathbf{1}_{\{Y_i \leq \theta\}} - \tau) K\left(\frac{X_i - x}{h}\right). \quad (\text{A1})$$

Let

$$\Psi(\theta; x) = E\left[\frac{\delta}{G(Y | X)} (\mathbf{1}_{\{Y \leq \theta\}} - \tau) K_h(X - x)\right] \quad (\text{A2})$$

denote the corresponding population estimating function.

Step 1: Identification of the population root.

Conditioning on $X = x$ and using the independent censoring assumption,

$$E\left[\frac{\delta}{G(Y | X)} \mathbf{1}_{\{Y \leq \theta\}} | X = x\right] = P(T \leq \theta | X = x). \quad (\text{A3})$$

Hence,

$$\Psi(\theta; x) = f_X(x)(F_{T|X}(\theta | x) - \tau) + o(1), \quad (\text{A4})$$

uniformly in a neighborhood of $Q_\tau(x)$.

By definition of the conditional quantile,

$$F_{T|X}(Q_\tau(x) | x) = \tau. \quad (\text{A5})$$

Assumption (A4) further implies that $f_{T|X}(Q_\tau(x) | x) > 0$, ensuring that $\Psi(\theta; x)$ is strictly increasing in θ and admits a unique zero at $\theta = Q_\tau(x)$.

Step 2: Uniform convergence of the empirical process.

Under Assumptions (A2)–(A5), standard results for kernel-weighted empirical processes yield

$$\sup_{\theta \in \Theta} |\Psi_n(\theta; x) - \Psi(\theta; x)| \xrightarrow{p} 0, \quad (\text{A6})$$

for any compact neighborhood Θ of $Q_\tau(x)$. The uniform consistency of $\widehat{G}$ guarantees that replacing G by $\widehat{G}$ does not affect the asymptotic behavior (Wang and Wang, 2009).

Step 3: Convergence of the estimator.

Since $\Psi(\theta; x)$ has a unique zero and $\Psi_n(\theta; x)$ converges uniformly to $\Psi(\theta; x)$, consistency of $\widehat{Q}_\tau(x)$ follows from the arg-zero theorem for Z-estimators (Qin and Lawless, 1994),

$$\widehat{Q}_\tau(x) = \arg\{\Psi_n(\theta; x) = 0\} \xrightarrow{p} Q_\tau(x), \quad (\text{A7})$$

which completes the proof. $\square$

A.2. Proof of Theorem 2 (Asymptotic Normality)

Proof. A first-order Taylor expansion of $\Psi_n(\theta; x)$ around $Q_\tau(x)$ gives

$$0 = \Psi_n(Q_\tau(x); x) + (\widehat{Q}_\tau(x) - Q_\tau(x)) \dot{\Psi}_n(Q_\tau(x); x) + o_p(|\widehat{Q}_\tau(x) - Q_\tau(x)|), \quad (\text{A8})$$

where $\dot{\Psi}_n(\theta; x)$ denotes the derivative with respect to θ .

Step 1: Asymptotic distribution of the score term.

By the central limit theorem for kernel-weighted IPCW estimators,

$$\sqrt{nh^d} \Psi_n(Q_\tau(x); x) \xrightarrow{d} \mathcal{N}(0, V(x)), \quad (\text{A9})$$

where

$$V(x) = \tau(1 - \tau)R(K)f_X(x)^{-1}, \quad (\text{A10})$$

Step 2: Convergence of the derivative term.

Standard kernel arguments imply

$$\dot{\Psi}_n(Q_\tau(x); x) \xrightarrow{p} f_X(x)f_{T|X}(Q_\tau(x) | x). \quad (\text{A11})$$

Step 3: Application of Slutsky's theorem.

Combining the above results,

$$\sqrt{nh^d}(\hat{Q}_\tau(x) - Q_\tau(x)) = -\frac{\sqrt{nh^d}\Psi_n(Q_\tau(x); x)}{\dot{\Psi}_n(Q_\tau(x); x)} + o_p(1), \quad (\text{A12})$$

which converges in distribution to the stated normal limit (Owen 2001). This completes the proof. $\square$

A.3. Proof of Theorem 3 (Wilks-Type Theorem)

Proof. The empirical likelihood is constructed subject to the constraint

$$\sum_{i=1}^n p_i \frac{\delta_i}{\hat{G}(Y_i | X_i)} (\mathbf{1}_{\{Y_i \leq Q_\tau(x)\}} - \tau) K_h(X_i - x) = 0. \quad (\text{A13})$$

Let λ_n denote the associated Lagrange multiplier.

A second-order Taylor expansion of the empirical likelihood ratio yields

$$-2 \log R(Q_\tau(x)) = nh^d \frac{\Psi_n^2(Q_\tau(x); x)}{\hat{V}_n(x)} + o_p(1), \quad (\text{A14})$$

where $\hat{V}_n(x)$ is a consistent estimator of $V(x)$.

Since

$$\sqrt{nh^d} \Psi_n(Q_\tau(x); x) \xrightarrow{d} \mathcal{N}(0, V(x)), \quad (\text{A15})$$

it follows that

$$-2 \log R(Q_\tau(x)) \xrightarrow{d} \chi_1^2, \quad (\text{A16})$$

which establishes the Wilks-type result. $\square$

Supplementary Materials

The supplementary materials include additional simulation results (Tables 1 and 2) and the R code used to reproduce the simulation study presented in this article.

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Data Availability Statement

The dataset analyzed in this study is publicly available in the R *survival* package. Specifically, the Primary Biliary Cirrhosis (PBC) data (pbc) was used, from which the variables `time`, `status`, `log(bilirubin)`, `albumin`, and `prothrombin` were extracted for analysis. Further information and documentation are available at: <https://cran.r-project.org/package=survival>.

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